Reg. No. | | | | | | | | | | |

[17EE33]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER–2018

ELECTRIC CIRCUIT ANALYSIS

Time:3 Hours **Maximum Marks:70**

Instructions: i. Answer any **FIVE** full Questions choosing atleast two from each part.

ii. Missing data may be suitably assumed.

PART – A

1. a. Find the power delivered by the sources in the network shown in fig.1a. Use mesh method. **06**

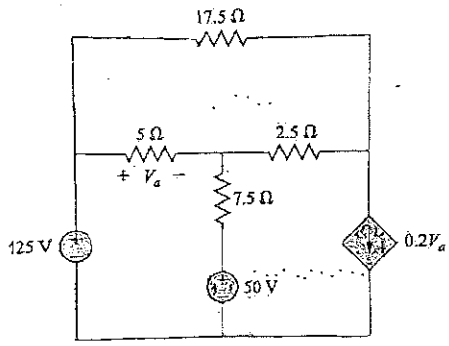


Fig 1(a)

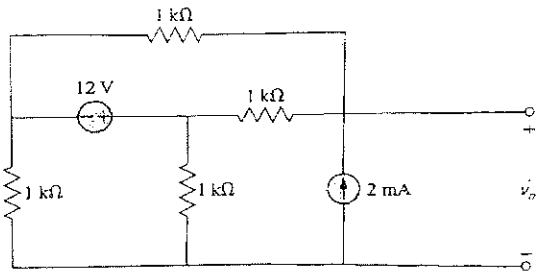


Fig 1(b)

b. Find the voltage V_o using super node technique in the network shown in fig.1b. **05**

c. A square based pyramid has its eight edges each representing a resistance of 30 Ohms. Find the equivalent resistance across the apex and one of the corners. **03**

2. a. Define the following terms: **03**

i. Tie-set,

ii. Cut-set schedule,

iii. Co-tree.

b. Construct the element node incidence matrix for the graph shown in fig 2b. **05**

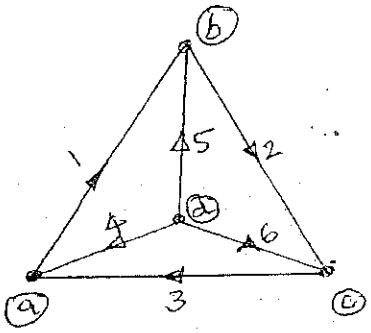


Fig. 2(b)

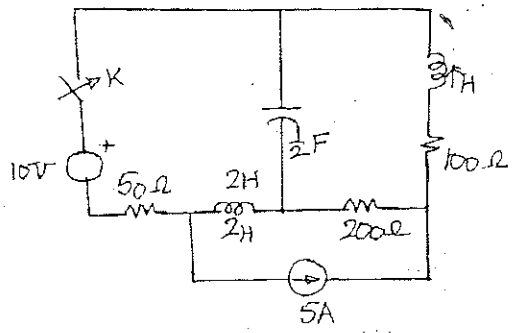


Fig. 2(c)

c. For the network shown in fig 2c., construct the dual network. **06**

3. a. State and explain Millman's theorem. 03

b. Verify the Reciprocity theorem for the network shown in fig.3b. 04

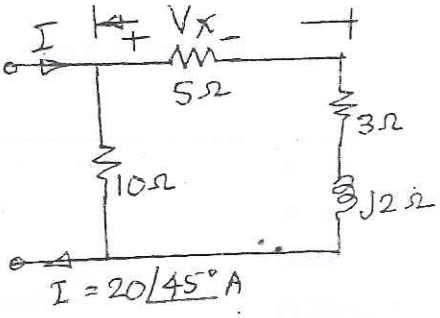


Fig. 3(b)

c. For the network shown in fig. 3c, obtain the Thevenin's and Norton's equivalent circuit across the output terminals A-B. 07

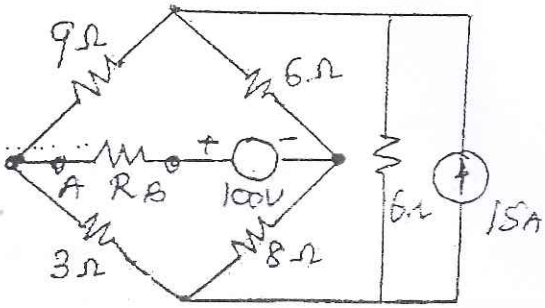


Fig. 3(c)

4. a. A fixed condenser is placed in parallel with a choke coil in which the resistance is fixed and the inductance is variable. Show that at resonance, 04

$$X_L = \frac{X_C}{2} \pm \sqrt{\{(\frac{X_C}{2})^2 - R^2\}}$$

b. Explain the construction of the locus of the total current of a parallel R-L and R-C branch circuit in which the inductance value is varied from zero to infinity 04

c. A two circuit parallel circuit consists of an R-L branch and an R-C branch having values of $R_L = 15\ \Omega$, $R_C = 30\ \Omega$, $X_C = 30\ \Omega$, $V = 120$ V, $f = 60$ Hz. X_L is variable. For the condition of resonance calculate the two values of inductance and the corresponding two values of total currents when connected across a source of 150 V, 50 Hz. 06

PART – B

5. a. A three phase unbalanced star connected load is excited by a balanced three phase supply. Derive an expression for the potential difference between the supply neutral and the load neutrals. Draw the vector diagram. 07

b. The currents in RY, YB and BR branches of a delta connected system with symmetrical voltages applied are 25 A at 0.8 power factor lagging, 30 A at 0.7 power factor leading and 20 A at unity power factor respectively. Determine the currents in each line. Phase sequence R-Y-B. 07

6. a. Explain the importance of initial conditions in networks and discuss the behavior of R, L and C at t=0+ and at steady state for the cases of with and without initial stored energy in L and C. 06

b. In the circuit shown in fig.6b, steady state is reached with switch K open. at t=0 sec, the switch is closed. Find i_1, i_2, Di_1 and Di_2 at t=0+. **08**

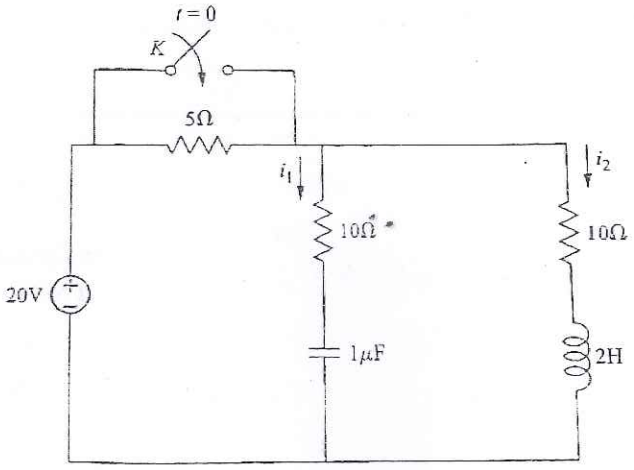


Fig. 6(b)

7. a. Show that the Laplace transformation of a unit impulse function is unity. **04**

b. State and prove Final value theorem.

c. For the periodic waveform shown in fig. write the expression for f(t) and find its Laplace Transformation.

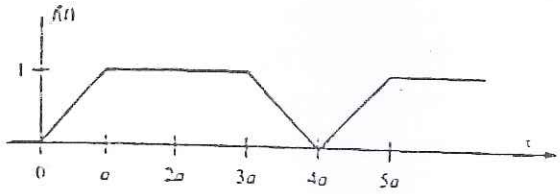


Fig. 7(c)

8. a. Using Laplace transformations find the ratio of Vo(s) to Vi(s) for the circuit shown in fig 8a if all the initial conditions are zero. **08**

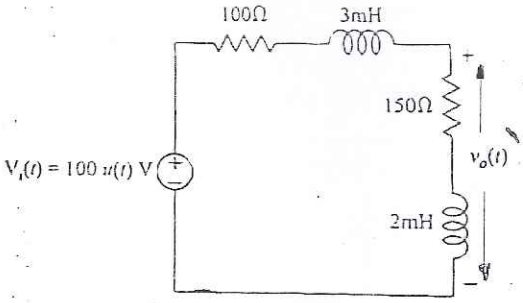
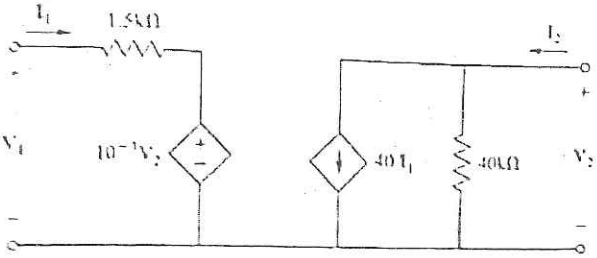


Fig.8a

Fig. 8(a)

b. For the network shown in figure 8c, obtain the T-parameters. **06**